SS-R-501D
June 20, 1969
SUPERSEDING
Int. Fed. Spec. SS-R-00501C(GSA-FSS)
June 6, 1967 and
Fed. Spec. SS-R-501A
September 28, 1948

FEDERAL SPECIFICATION

ROOFING FELT, ASPHALT-PREPARED, SMOOTH SURFACED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers smooth surfaced organic roofing felt saturated and coated with asphaltic materials and surfaced with fine mineral matter.

1.2 Classification. The roofing felt shall be of the following types, as specified (see 6.2).

Type I - ~~37~~ 35 pounds
Type II - ~~39~~ 45 pounds
Type III - 55 pounds
Type IV - 65 pounds

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

NN-P-71 - Pallets, Material Handling, Wood, Double Faced, Stringer Construction.
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-636 - Box, Fiberboard.
PPP-S-760 - Strapping, Nonmetallic, (And Connectors).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-R-3423 - Roofing Felt, In Rolls, Packaging of.

FSC 5650

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A239 - Method of Test for Uniformity of Coating by the Preece Test (Copper Sulphate Dip) on Zinc-Coated (Galvanized) Iron or Steel Articles.
D224 - Specifications for Asphalt Roll Roofing Surfaced with Powdered Talc or Mica.
D228 - Methods of Testing Asphalt Roll Roofing, Cap Sheets, and Shingles.
D2626 - Specification for Asphalt-Base Sheet for Use in Construction of Built-Up Roofs.

E96 - Methods of Test for Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

National Classification Board:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P St., N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Types I and II. Types I and II roofing felts shall be in accordance with ASTM D2626 except for sampling, rejection, packaging, packing, and marking and the requirements specified herein (see 4.4.2).

3.2 Types III and IV. Types III and IV roofing felts shall be in accordance with ASTM D224 except for sampling, rejection, packaging, packing, and marking and the requirements specified herein (see 4.4.2).

3.3 Surface finish, types III and IV. In lieu of talc or mica, the use of other fine mineral surfacing shall be permitted on the weather side of types III and IV. roofing felts.

3.4 Water vapor transmission. When specified (see 6.2), types III and IV roofing felt shall have a maximum water vapor transmission (WVT) rate of 0.3 perm when tested in accordance with 4.4.3.

3.5 Nails and lap-cement. When specified (see 6.2), nails and lap-cement shall be furnished with each roll of roofing felt.

3.5.1 Nails. The gage of the wire shall be not greater than Washburn and Moen No. 10 (0.135 inch) nor less than Washburn and Moen No. 12 (0.105 inch). The head shall be not less than 3/8 inch in diameter, nor less than 0.025 inch in thickness. The shank shall be not less than 3/4 inch in length and shall be smooth, barbed, or square and twisted (spiral). If barbed, the barbing shall not increase the diameter of the lower two-thirds more than 0.01 inch. The lower end of the shank shall be

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pointed. If square and twisted (spiral), the cross-sectional area shall be equivalent to that specified above. Nails shall be galvanized or sherardized, and shall resist four one-minute dips in a copper sulfate solution when tested in accordance with ASTM A239. An average of not less than 252 nails shall be furnished per hundred square feet of roofing felt.

3.5.2 Lap-cement. The cement shall be composed of asphaltic materials dissolved in a volatile solvent, and shall be of such a nature as to firmly bind the laps without injurious effect upon the roofing. Not less than 3/4 pint of cement shall be furnished with each 100 square feet of roofing felt. The addition of 5 to 20 percent by weight of short-fibered asbestos shall be permitted provided the quantity of cement per 100 square feet of roofing felt is increased by 1/4 pint.

3.6 Rolls. No roll of roofing shall contain more than two pieces nor shall there be more than 3 percent of rolls in a lot containing two pieces. Rolls shall be tightly wound, shall not be flattened, and ends of rolls shall not be battered.

3.7 Workmanship. The finished roofing felt shall be free of visible defects such as holes, ragged or untrue edges, breaks, cracks, tears, protuberances, and indentations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all asphalt roofing rolls of the same type offered for delivery at one time.

4.2.2 Dimensional and visual examination of end item. Sampling for these characteristics shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Samples for tests. Sampling for tests shall be in accordance with level S-2 in MIL-STD-105.

4.3 Examination.

4.3.1 Dimensional and visual examination. Roofing felt selected in accordance with 4.2.2 shall be examined and measured for defects listed in table I. The Acceptable Quality Levels (AQL's) for defects shall be 2.5 percent defective for major defects, and 4.0 percent defective for minor defects. The roofing felt shall be measured with a rule graduated in 1/32 inch increments. Dimensions shall be determined in accordance in accordance with ASTM D228.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Type not as specified.	X	
Coating and surfacing not uniformly applied up to edges of sheet.		X
Width not within tolerance.	X	
Average area of rolls less than specified.		X
Nails or cement not as specified.	X	
Roll contains more than two pieces or more than 3 percent of rolls contains two pieces.	X	
Rolls not tightly wound.		X
Rolls flattened, unusable.	X	
Ends of rolls battered, unusable.	X	
Rolls stick, tear, or crack while unrolling.	X	
Holes, tears, breaks, or cracks in felt.	X	
Not free of protuberances or indentations.		X
Ragged or untrue edges.	X	
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability.		X

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4.3.2 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table II. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers full prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 percent defective.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as loose strapping, inadequate stapling or improper stacking.

4.4 Test methods.

4.4.1 Acceptable quality level. The AQL for tests shall be 4.0 percent defective.

4.4.2 Pliability, behavior on heating, breaking strength, and weights. Pliability, behavior on heating, breaking strength, and weights of roofing felt shall be determined in accordance with ASTM D228 (see 3.1 and 3.2).

4.4.3 Water vapor transmission. Water vapor transmission shall be determined in accordance with procedure B in ASTM E96 (see 3.4).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The rolls of roofing felt shall be packaged as specified in MIL-R-3423.

5.1.2 Level B, civil agencies (see 6.5). Each roll of roofing felt shall be packaged as specified for level A except the wrapping paper shall conform to UU-P-268, grade B. Alternatively, each roll may be packaged in a close fitting box conforming to PPP-B-636, class domestic, in lieu of wrap. The box shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C. Packaging shall be in accordance with the level C requirements of MIL-R-3423.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The rolls of roofing felt shall be packed as specified in MIL-R-3423.

5.2.2 Level B. The rolls, packaged as specified in 5.1, arranged on end shall be palletized on a pallet conforming to NN-P-71, Type I or II, except the maximum length and width shall be 46 inches by 52 inches. The rolls shall be circumferentially secured with nonmetallic strapping conforming to PPP-S-760, Type II or III, with a minimum breaking strength of 680 pounds. Alternatively, the packaged rolls arranged on end shall be palletized on an expendable wooden pallet as described in 5.2.2.1. The top corners of the rolls along the width of the pallet shall be protected by a full length V board as described in 5.2.2.2. The rolls shall be secured circumferentially and girthwise with nonmetallic strapping conforming to PPP-S-760, type II or III, with a minimum breaking strength of 680 pounds. The circumferential strap will be approximately at the center of the roll length. The girthwise strap shall be around the length of the pallet and V boards and as near the center of the load as the center stringer will permit.

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5.2.2.1 Expendable wooden pallet. Expendable wooden pallets shall be a maximum of 46 inches long and 52 inches wide. Deckboards and stringers shall be Douglas Fir, standard grade or better. Each deckboard or stringer must be free of decay and any other defect which would render it unfit for satisfactory service in a pallet. Boards may be S1S or S2S, however, thickness must be uniform within 1/16 inch and minimum width must be within 1/2 inch of width specified. Overall pallet height must be uniform within 1/8 inch. Pallets must be square. Deckboards must be evenly spaced and not over 2-5/8 inches apart. The deckboards shall be a minimum of 1 by 6 inches nominal. The stringers shall be a minimum of 2 by 4 inches nominal. The deckboards shall be nailed to the stringers at all contact points with a minimum of two cement coated 8d box nails. One stringer shall be nailed along the center of the deckboards. The two outer stringers will be set in approximately 3 inches from the ends of the deckboard.

5.2.2.2 V Boards. The V boards shall be constructed of one 1 by 6 inches nominal and one 2 by 2 inches nominal Douglas Fir. standard grade or better. The length shall be less than the width of the pallet but long enough to protect all rolls in the outer line.

The 1 by 6 shall be nailed to the 2 by 2 in such a manner as to form a V, with a minimum of four 8d cement coated box nails equally spaced.

5.2.3 Level C. The roofing felt packaged as specified in 5.1 shall be packed to assure carrier acceptance and safe delivery at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. In addition to any marking required by the contract or order the rolls and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129 as applicable.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II. Types I and II roofing felts are intended for use as the first ply of built-up roofing. It may also be used for temporary roofing or temporary flashings, as an insulation vapor barrier, as a vapor barrier under concrete slabs, or as a crawl space cover.

6.1.2 Types III and IV. Types III and IV roofing felts are intended for use as a single layer roofing, as a cap sheet in built-up roofing, as a vapor barrier under concrete slabs, or as a crawl space cover.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type of roofing felt (see 1.2).
(c) When water vapor transmission requirements are applicable to types III and IV roofing felts (see 3.4).
(d) When nails and lap-cement are required (see 3.5).
(e) Size of lot, if different from 4.2.1.
(f) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Square. A square is defined by the roofing industry as one hundred (100) square feet of roof area. Rolls of roofing are available in units of 1, 1-1/2, or 2 square rolls, which will cover 100, 150, or 200 square feet, respectively, of roof surface.

6.4 Area of coverage.

6.4.1 108 square feet. When a 2 inch lapping edge is used, 108 square feet of roofing felt is required to cover 100 square feet of roof surface.

6.4.2 114 square feet. When a 4 inch lapping edge is used, 114 square feet of roofing felt is required to cover 100 square feet of roof surface.

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6.5 Level B packaging (see 5.1.2) has been developed by GSA for civil procurement. For Navy procurement, details of level B packaging, when required, should be developed at the time of procurement, when all conditions are known.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Army - CE

User activity:

Navy - MC

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA-FSS

Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

SS-R-501D ROOFING FELT, ASPHALT-PREPARED, SMOOTH SURFACED

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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